

Biology Mid-Term Examination

Grade 10 | Total Marks: 50

Time: 2 Hours

1. 2 marks
Which blood vessel carries blood away from the heart?
2. 2 marks
Which of the following organelles is primarily involved in photosynthesis? (a) Mitochondria (b) Ribosome (c) Chloroplast (d) Nucleus
3. 2 marks
Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outlining the two major stages.
4. 2 marks
Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta; include valve names.
5. 2 marks
Draw a labelled diagram of an alveolus showing capillaries and air space. Label: alveolar sac, capillary, direction of gas exchange.
6. 5 marks
Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine, liver, pancreas).
7. 5 marks
Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule, collecting duct).
8. 5 marks
Explain the structural differences between palisade mesophyll and spongy mesophyll and state how each aids its function.
9. 5 marks
Describe transpiration in plants in 2-3 sentences and name two environmental factors that increase its rate.
10. 5 marks
Explain how xylem vessel structure facilitates water transport (mention one structural feature and its role).
- 11a. 2 marks
A diagram shows two potted plants — Plant A in bright light with broad green leaves, Plant B in dim light with pale elongated leaves.